Quantum Mechanics

Lecture 10: The Quantum Harmonic Oscillator and Its Ladder of States

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Chapter 1

The Quantum Harmonic Oscillator and Its Ladder of States

1.1 The Universal Model Near Stable Equilibrium

This is the last lecture in the present sequence, though not the last word on quantum mechanics. Angular momentum is the other natural closing subject, but the harmonic oscillator fills the available time and angular momentum will have to wait for the later discussion of quantum fields.

The oscillator deserves that time because it is not merely one special spring problem. Whenever a system has a stable equilibrium and is displaced slightly from it, the first nontrivial approximation to its motion is usually harmonic. A weight suspended from a spring is the obvious model. Molecular vibrations, sound modes, electromagnetic modes, and small disturbances of many more complicated systems share the same mathematics.

Consider the hanging mass. Gravity pulls downward with an approximately constant force, while the spring force grows with extension. At one position the two balance. If x measures displacement from that equilibrium rather than from the spring's unstretched length, gravity disappears from the equation of small motion and the restoring force is

$$F(x) = -kx. \tag{1.1}$$

The minus sign says that either displacement produces a force back toward $x = 0$.

The same conclusion follows for a general potential $V(q)$ with a nondegenerate minimum at $q = q_0$. Put $x = q - q_0$. Taylor expansion gives

$$V(q_0 + x) = V(q_0) + \underbrace{V'(q_0)}_{0}\,x + \frac{1}{2}V''(q_0)x^2 + O(x^3). \tag{1.2}$$

The constant does not affect the force, the linear term vanishes at the minimum, and the positive quadratic coefficient is the spring constant: $k = V''(q_0) > 0$. Harmonic motion is therefore the local language of stable equilibrium.

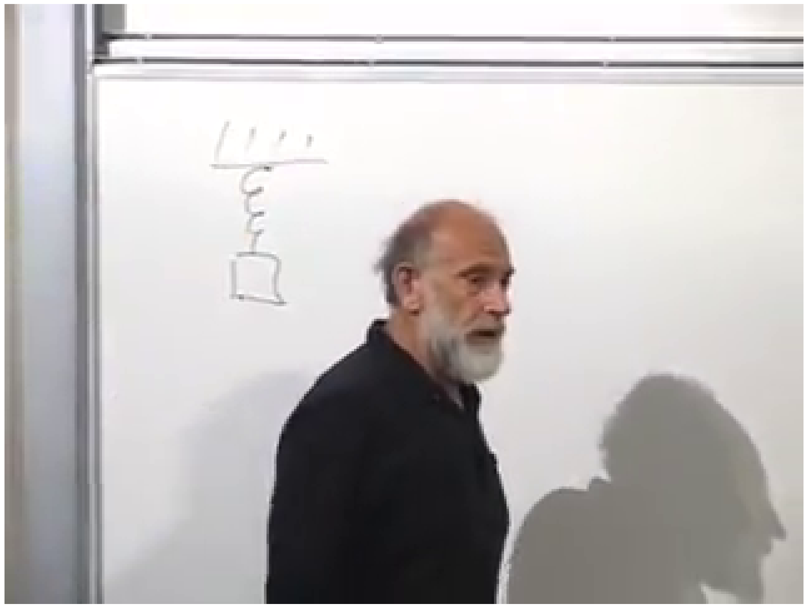

Figure 1.1: A suspended mass and spring provide the concrete oscillator model; the coordinate is measured from the force-balanced equilibrium.

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1.2 Classical Preparation

Before quantizing, let us put the classical system into a convenient form. For mass m and angular frequency $\omega = \sqrt{k/m}$,

$$L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}m\omega^2 q^2. \tag{1.3}$$

The rescaling $x = \sqrt{m}\,q$ absorbs the mass:

$$L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2. \tag{1.4}$$

We shall use this unit-mass coordinate through most of the lecture and restore m whenever its physical role matters.

The Euler–Lagrange equation gives

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}, \tag{1.5}$$

or

$$\boxed{\ddot{x} = -\omega^2 x.} \tag{1.6}$$

Its general real solution is

$$x(t) = A\cos\omega t + B\sin\omega t. \tag{1.7}$$

Twice differentiating either sinusoid returns the function multiplied by $-\omega^2$, so the frequency in the potential is exactly the frequency of the motion.

The canonical momentum is

$$p = \frac{\partial L}{\partial \dot{x}} = \dot{x}, \tag{1.8}$$

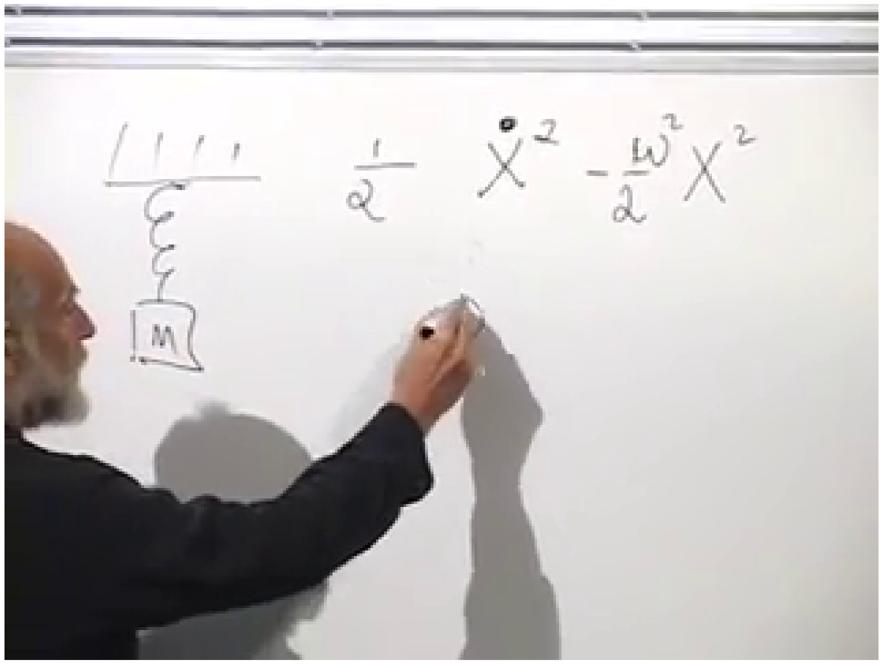

Figure 1.2: The spring model and its unit-mass Lagrangian, with a quadratic potential characterized by ω^2.

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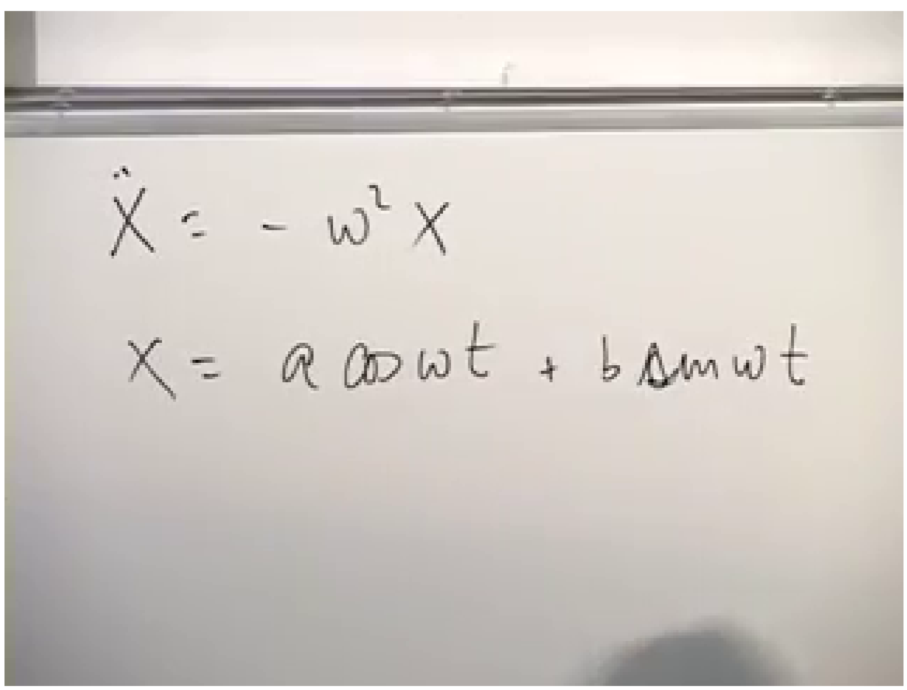

Figure 1.3: The restoring-force equation and its sinusoidal classical solution.

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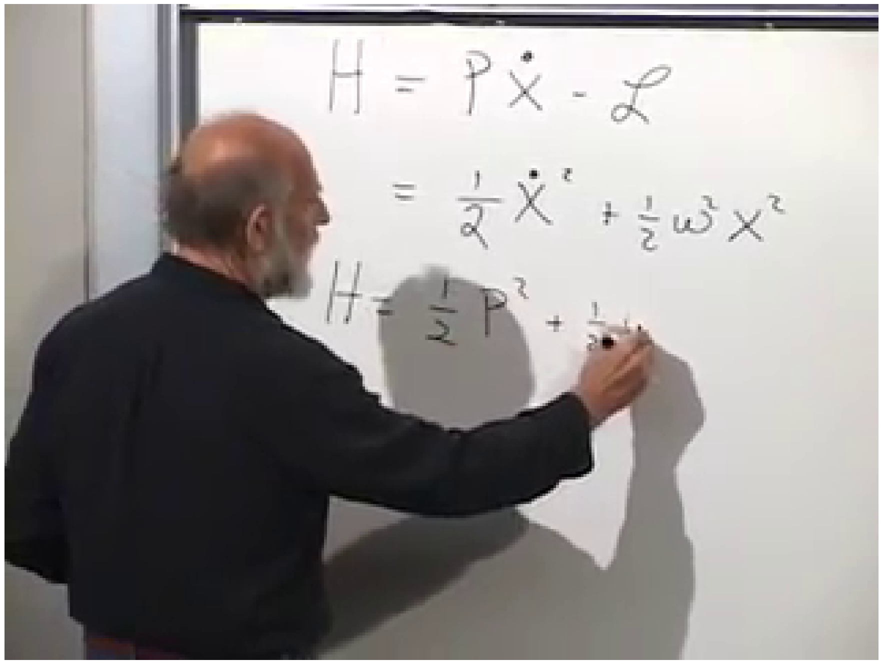

Figure 1.4: The Legendre transform rewrites the oscillator energy in the canonical variables x and p.
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and the Legendre transform produces

$$H = p\dot{x} - L \tag{1.9}$$
$$= \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2. \tag{1.10}$$

The Lagrangian is kinetic minus potential energy; the Hamiltonian is their sum.

1.3 Quantization and the Two Jobs of the Hamiltonian

The quantum state space consists of square-integrable wave functions $\psi(x)$. At a specified time,

$$|\psi(x)|^2 \, dx \tag{1.11}$$

is the probability of finding the oscillator coordinate in an interval dx, and normalization requires

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = 1. \tag{1.12}$$

In the position representation,

$$(\hat{x}\psi)(x) = x\psi(x), \qquad (\hat{p}\psi)(x) = -i\hbar\frac{\partial \psi}{\partial x}. \tag{1.13}$$

Writing $\hat{x}\,|\psi\rangle$ and writing multiplication by x are not two different operations; the second is the position-space representation of the first. Similarly, $\hat{p}^{\,2}$ means applying the momentum operator twice.

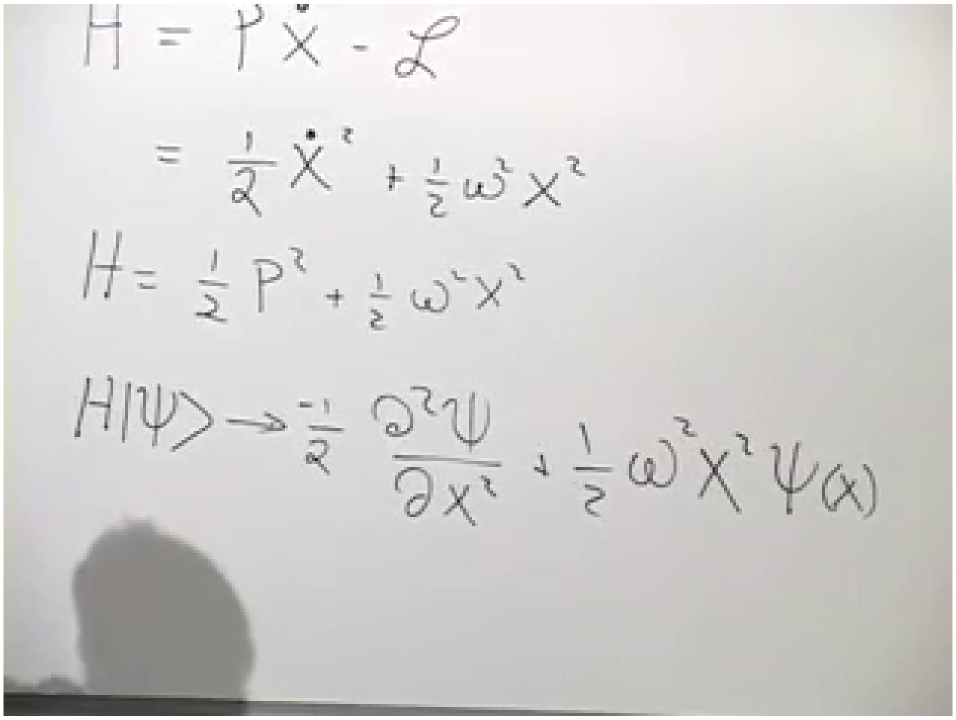

Figure 1.5: The classical Hamiltonian and its action as a differential operator on the wave function. *Lecture frame: 00:22:35.*

Promoting the classical variables in Equation (1.10) to operators gives

$$\hat{H} = \frac{1}{2}\hat{p}^{\,2} + \frac{1}{2}\omega^2\hat{x}^{\,2}. \tag{1.14}$$

Its action on a wave function is

$$(\hat{H}\psi)(x) = -\frac{\hbar^2}{2}\frac{\partial^2\psi}{\partial x^2} + \frac{1}{2}\omega^2 x^2\psi(x). \tag{1.15}$$

The Hamiltonian has two closely related jobs. First, it generates time evolution:

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2}\frac{\partial^2}{\partial x^2} + \frac{1}{2}\omega^2 x^2\right]\psi(x,t). \tag{1.16}$$

Even if $\psi(x,0)$ happens to be real, the factor i generally produces an imaginary component immediately. A nonstationary wave function is therefore intrinsically complex.

Question from the audience. Why are the derivatives partial derivatives rather than ordinary derivatives? [a]

Response. The wave function depends independently on x and t. The derivative $\partial_t\psi$ changes time while holding position fixed, whereas $\partial_x\psi$ changes position while holding time fixed. Time labels evolution; it does not have the same operator status as position in this formulation.

[a] Lecture timestamp: 00:21:24–00:24:07.

Given an initial wave function, one could integrate Equation (1.16) forward in small time steps. The second job of $\hat{H}$, however, gives a more revealing route. Solve the stationary eigenvalue problem

$$\hat{H}\psi_E = E\psi_E, \tag{1.17}$$

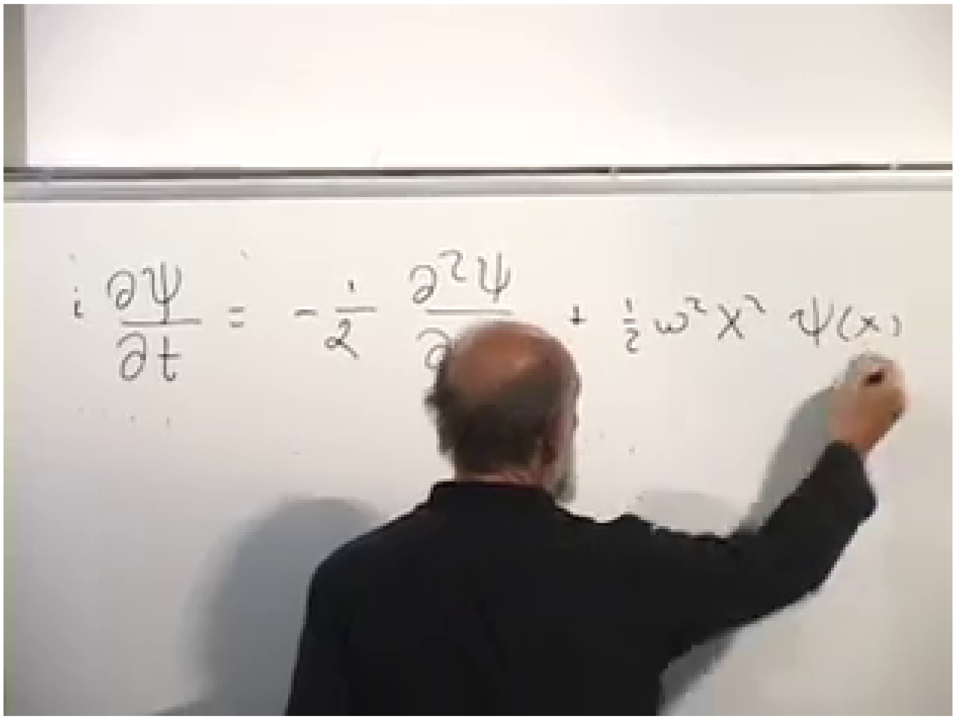

Figure 1.6: The time-dependent Schrödinger equation for the oscillator.
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or explicitly,

$$-\frac{\hbar^2}{2}\frac{d^2\psi_E}{dx^2}+\frac{1}{2}\omega^2x^2\psi_E=E\psi_E. \tag{1.18}$$

Once the eigenfunctions are known, arbitrary initial data may be expanded as

$$\psi(x,0)=\sum_n c_n\psi_n(x), \tag{1.19}$$

and its time evolution is immediate:

$$\psi(x,t)=\sum_n c_n e^{-iE_nt/\hbar}\psi_n(x). \tag{1.20}$$

1.4 Why the Bound-State Energies Are Discrete

Equation (1.18) is second order, so for a fixed E it has two linearly independent local solutions. Far from the origin, the quadratic potential dominates. The two asymptotic behaviors are of the form

$$\psi(x)\sim\exp\left(\pm\frac{\omega x^2}{2\hbar}\right). \tag{1.21}$$

One decays and one grows. For a generic trial energy, a solution adjusted to decay at $+\infty$ contains a growing component at $-\infty$, or vice versa. Only selected energies allow the growing component to vanish at both ends.

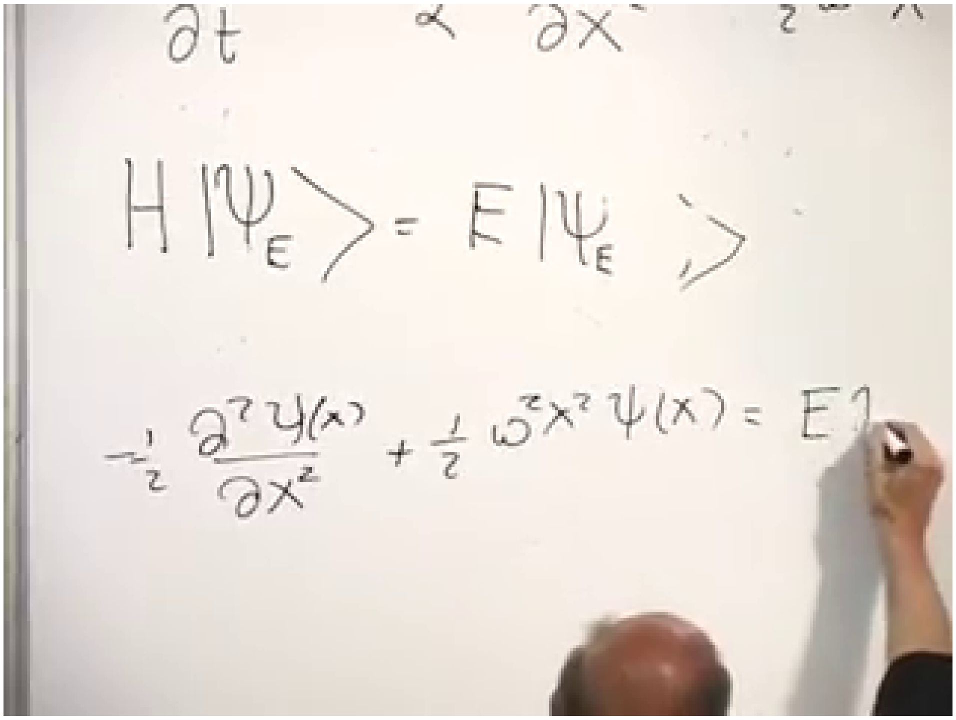

Figure 1.7: The stationary Schrödinger equation simultaneously determines the oscillator eigenfunctions and their allowed energies.

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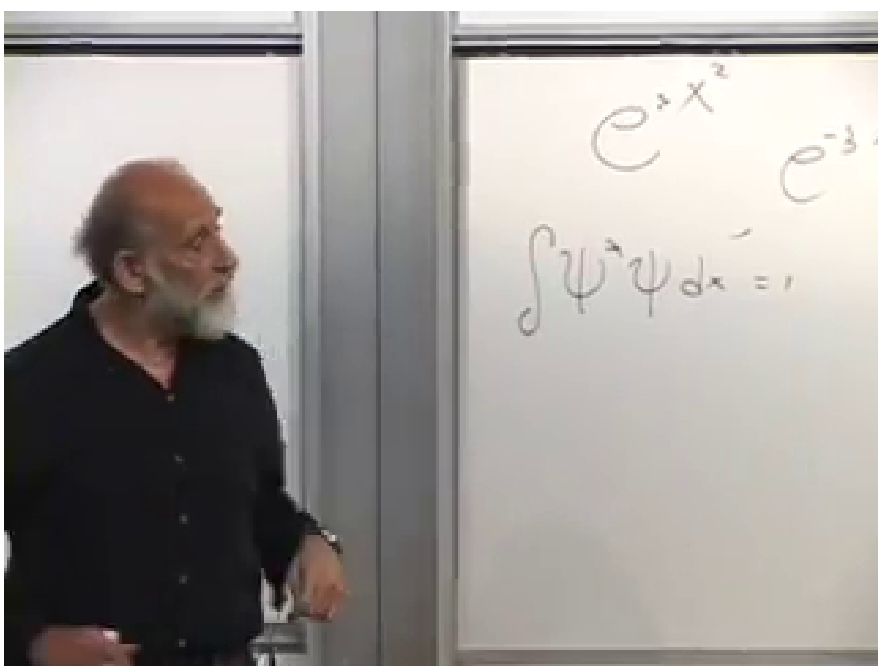

Figure 1.8: Normalizability rejects the exponentially growing asymptotic branch and selects isolated energies that decay at both ends.

Lecture frame: 00:34:20.

This is the origin of the discrete bound-state spectrum. The differential equation exists for arbitrary E, but a physical state must belong to the Hilbert space. A function that grows exponentially cannot satisfy Equation (1.12); in the oscillator potential it also has divergent potential-energy expectation.

There is a second way to anticipate the lowest energy. Classically one minimizes

$$H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2 \tag{1.22}$$

by setting $x = p = 0$. Quantum mechanically,

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}, \tag{1.23}$$

so no normalized state can make both contributions vanish. The ground-state energy must be positive.

1.5 Finding the Gaussian Ground State

The asymptotic form suggests a Gaussian:

$$\psi_0(x) = C \exp\left(-\frac{\omega x^2}{2\hbar}\right), \tag{1.24}$$

where $\omega > 0$ denotes the positive oscillator frequency. Its first two derivatives are

$$\psi_0' = -\frac{\omega x}{\hbar}\psi_0, \tag{1.25}$$

$$\psi_0'' = \left(\frac{\omega^2 x^2}{\hbar^2} - \frac{\omega}{\hbar}\right)\psi_0. \tag{1.26}$$

Applying the Hamiltonian,

$$\hat{H}\psi_0 = -\frac{\hbar^2}{2}\left(\frac{\omega^2 x^2}{\hbar^2} - \frac{\omega}{\hbar}\right)\psi_0 + \frac{1}{2}\omega^2 x^2 \psi_0 \tag{1.27}$$

$$= \frac{1}{2}\hbar\omega\,\psi_0. \tag{1.28}$$

The two terms proportional to x^2 cancel, leaving

$$\boxed{E_0 = \frac{1}{2}\hbar\omega.} \tag{1.29}$$

Normalization fixes

$$C = \left(\frac{\omega}{\pi\hbar}\right)^{1/4}, \tag{1.30}$$

up to an irrelevant overall phase. For a physical coordinate q with mass m, the replacements $x = \sqrt{m}\,q$ give

$$\psi_0(q) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega q^2}{2\hbar}\right). \tag{1.31}$$

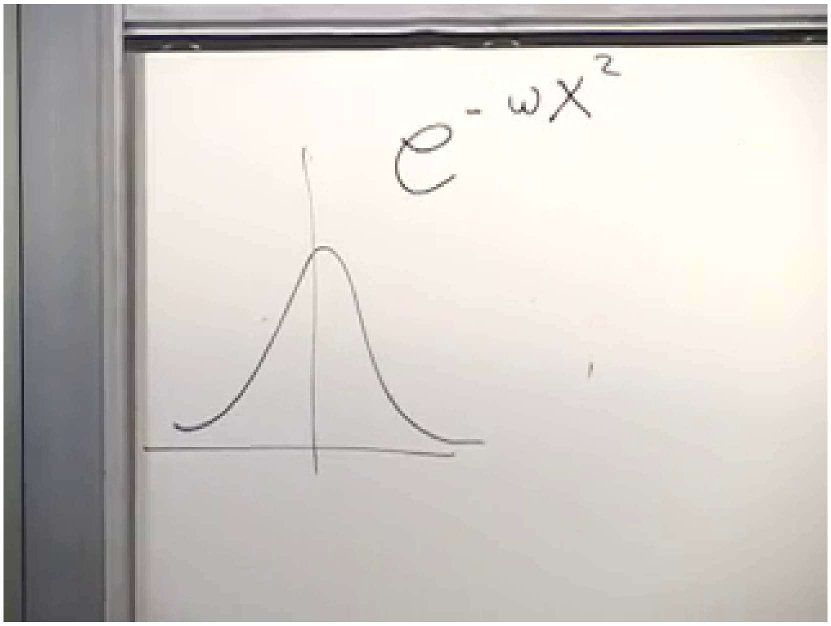

Figure 1.9: The Gaussian trial state is concentrated near the stable equilibrium and decays in both directions.

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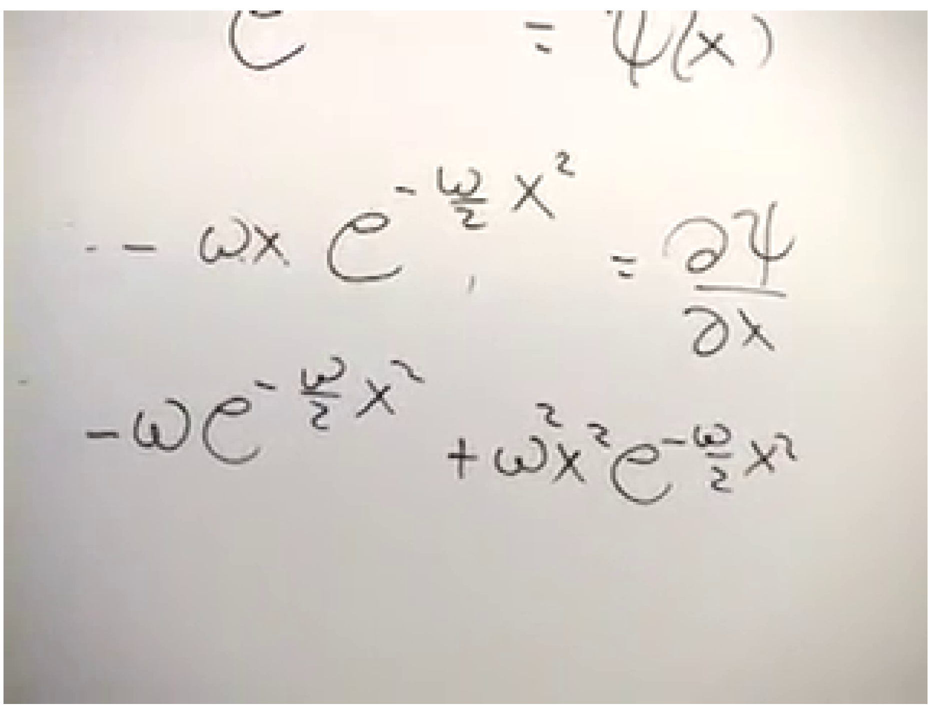

Figure 1.10: Differentiating the Gaussian twice produces a constant term and a quadratic term.

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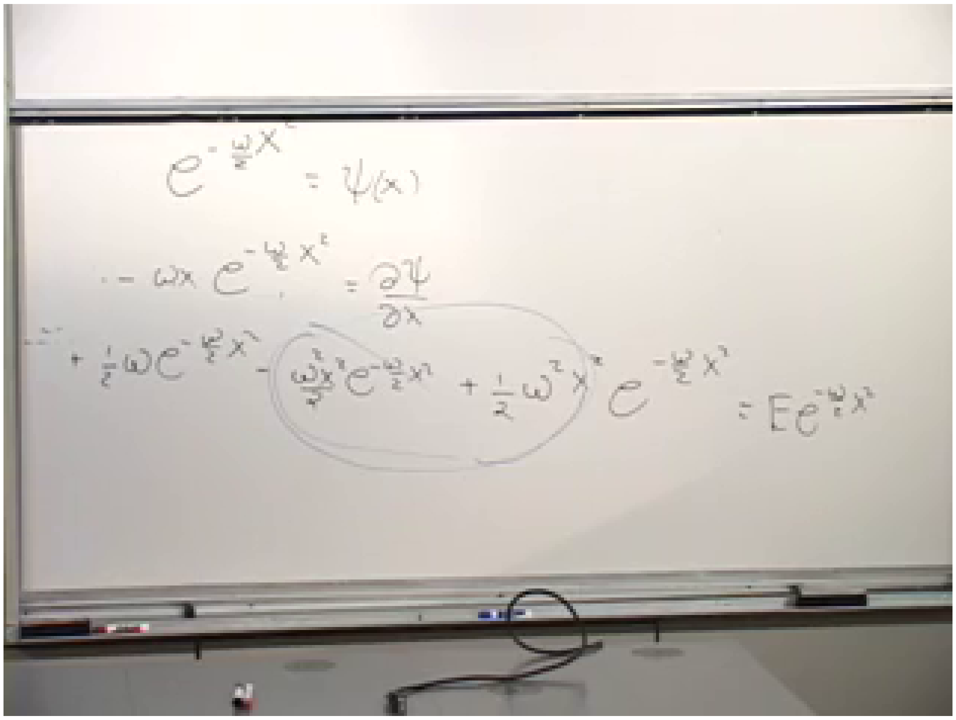

Figure 1.11: The Gaussian eigenfunction calculation leaves the zero-point energy after the quadratic terms cancel.

Lecture frame: 00:44:34.

Question from the audience. Does the uncertainty principle mean that the oscillator can never have zero energy? [a]

Response. Yes. A zero-energy state would require both positive terms $\langle p^2\rangle/2$ and $\omega^2\langle x^2\rangle/2$ to vanish. That would make both position and momentum sharp at zero, contradicting Equation (1.23). The lowest allowed value is $\hbar\omega/2$.

[a]Lecture timestamp: 00:45:23–00:46:13.

Question from the audience. Why not take $\psi(x) = 1$ as a zero-energy state? [a]

Response. A constant wave function is a formal zero-momentum eigenfunction, so the kinetic term vanishes, but it is neither normalizable on the line nor localized at $x = 0$. The potential term produces $\frac{1}{2}\omega^2 x^2 \psi$, which is not zero and is not proportional to the constant state. It therefore does not solve the oscillator eigenvalue equation.

[a]Lecture timestamp: 00:46:13–00:47:49.

Question from the audience. For $E = 0$, can some linear combination of the two second-order solutions remain bounded at both ends? [a]

Response. No. Initial data can be tuned so the solution decays in one direction, but at $E = 0$ it grows in the other. The exceptional tuning that removes the growing branch at both ends occurs only at the discrete eigenvalues, the lowest of which is $\hbar\omega/2$.

[a]Lecture timestamp: 00:48:02–00:50:27.

Question from the audience. Why must the ω in the Gaussian exponent be the positive root? [a]

Response. The classical equation contains only ω^2, so its sign is irrelevant there. In the wave function, $\exp[-\omega x^2/(2\hbar)]$ decays only for $\omega > 0$. Choosing the positive frequency is therefore the normalizable branch.

[a] Lecture timestamp: 00:50:33–00:51:15.

Restoring $\hbar$ is essential. Frequency has units of inverse time, while energy requires $\hbar\omega$. For a macroscopic spring the spacing is usually far too small to notice, and the zero-point energy disappears relative to ordinary classical scales as $\hbar \to 0$.

1.6 Factorizing the Hamiltonian

The Gaussian gives one eigenstate. To find all of them, factor the sum of squares. Define the unnormalized operators

$$\hat{b}_- = \hat{p} - i\omega\hat{x}, \qquad \hat{b}_+ = \hat{p} + i\omega\hat{x} = \hat{b}_-^\dagger. \tag{1.32}$$

Classically, $(p + i\omega x)(p - i\omega x) = p^2 + \omega^2 x^2$. For noncommuting quantum operators, the chosen order contributes an additional term:

$$\begin{aligned} \hat{b}_+\hat{b}_- &= (\hat{p} + i\omega\hat{x})(\hat{p} - i\omega\hat{x}) && (1.33)\\ &= \hat{p}^{\,2} + \omega^2\hat{x}^{\,2} + i\omega[\hat{x}, \hat{p}] && (1.34)\\ &= \hat{p}^{\,2} + \omega^2\hat{x}^{\,2} - \hbar\omega. && (1.35) \end{aligned}$$

Hence

$$\boxed{\hat{H} = \frac{1}{2}\hat{b}_+\hat{b}_- + \frac{1}{2}\hbar\omega.} \tag{1.36}$$

The first-order operator $\hat{b}_-$ annihilates the Gaussian:

$$\begin{aligned} \hat{b}_-\psi_0 &= \left(-i\hbar\frac{d}{dx} - i\omega x\right)\psi_0 && (1.37)\\ &= (i\omega x - i\omega x)\,\psi_0 = 0. && (1.38) \end{aligned}$$

Therefore the product $\hat{b}_+\hat{b}_-$ contributes zero on the ground state, and Equation (1.36) immediately returns $E_0 = \hbar\omega/2$.

Normalize the operators:

$$\hat{a} = \frac{\hat{b}_-}{\sqrt{2\hbar\omega}}, \qquad \hat{a}^\dagger = \frac{\hat{b}_+}{\sqrt{2\hbar\omega}}. \tag{1.39}$$

Since

$$[\hat{b}_-, \hat{b}_+] = 2\hbar\omega, \tag{1.40}$$

we obtain

$$\boxed{[\hat{a}, \hat{a}^\dagger] = 1.} \tag{1.41}$$

Equivalently, $[\hat{a}^\dagger, \hat{a}] = -1$. This resolves the sign uncertainty that arises if the order of the two operators is changed while they are being named.

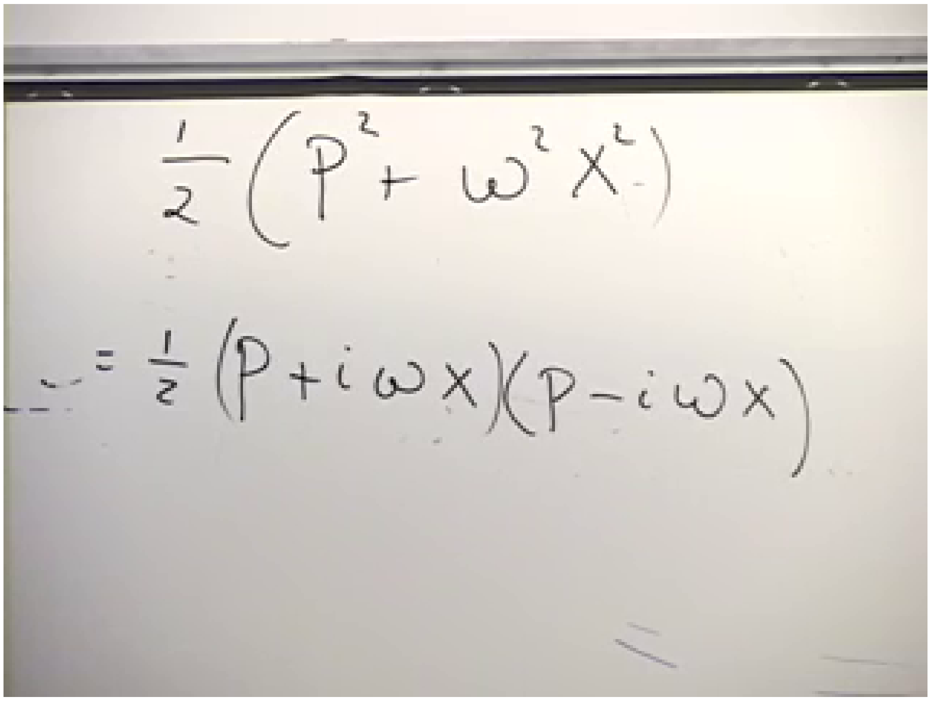

Figure 1.12: The classical-looking factorization must retain operator order in quantum mechanics.

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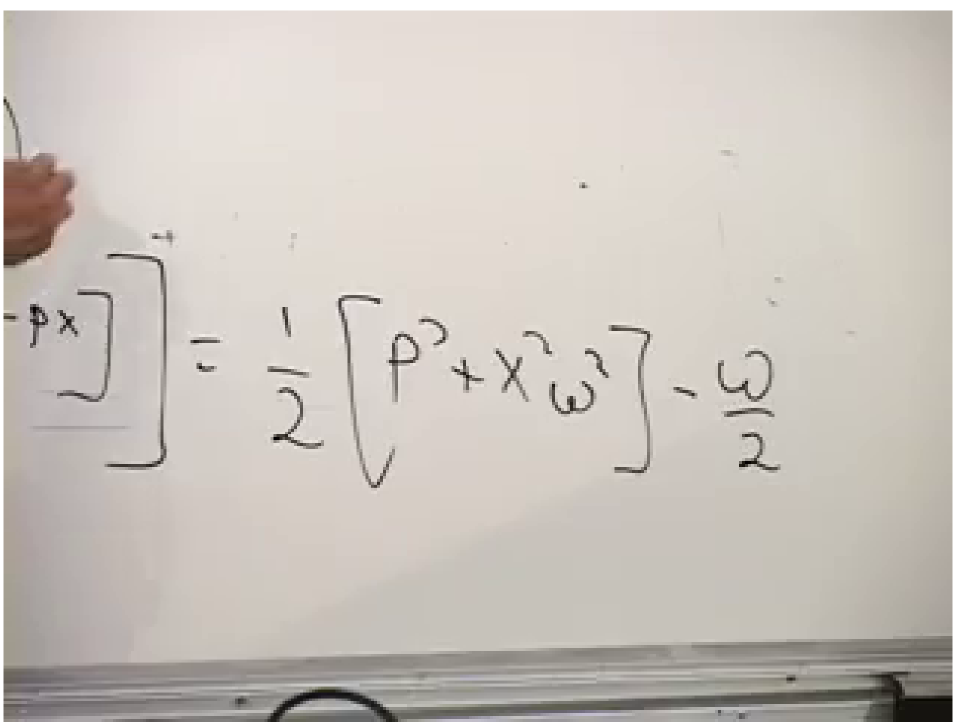

Figure 1.13: The noncommuting cross terms generate the additive zero-point contribution in the factorized Hamiltonian.

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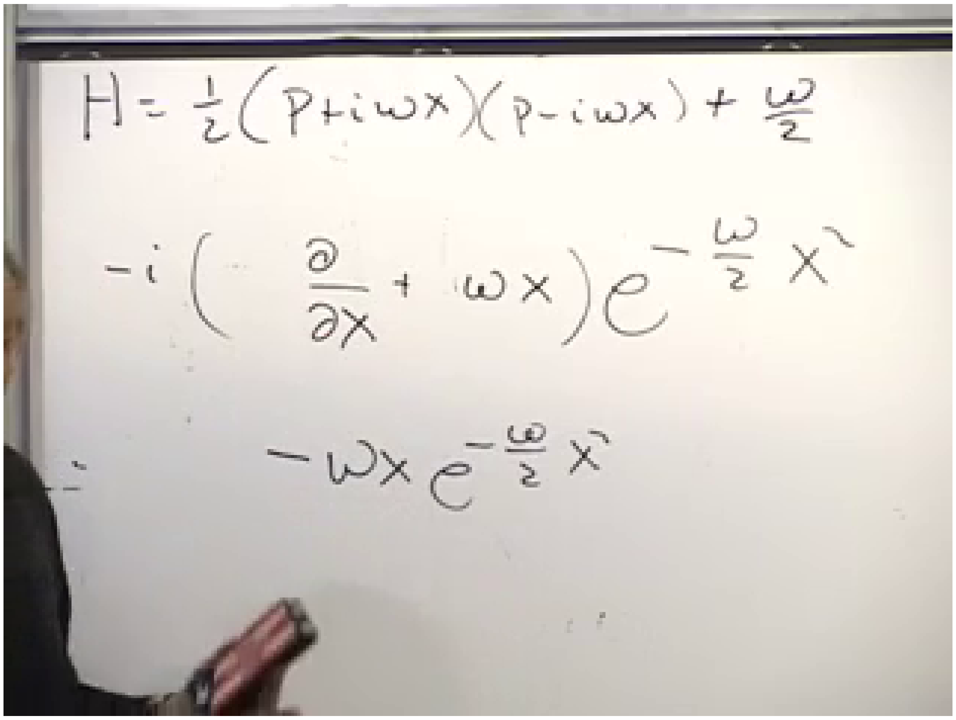

Figure 1.14: The lower first-order factor annihilates the Gaussian ground state.

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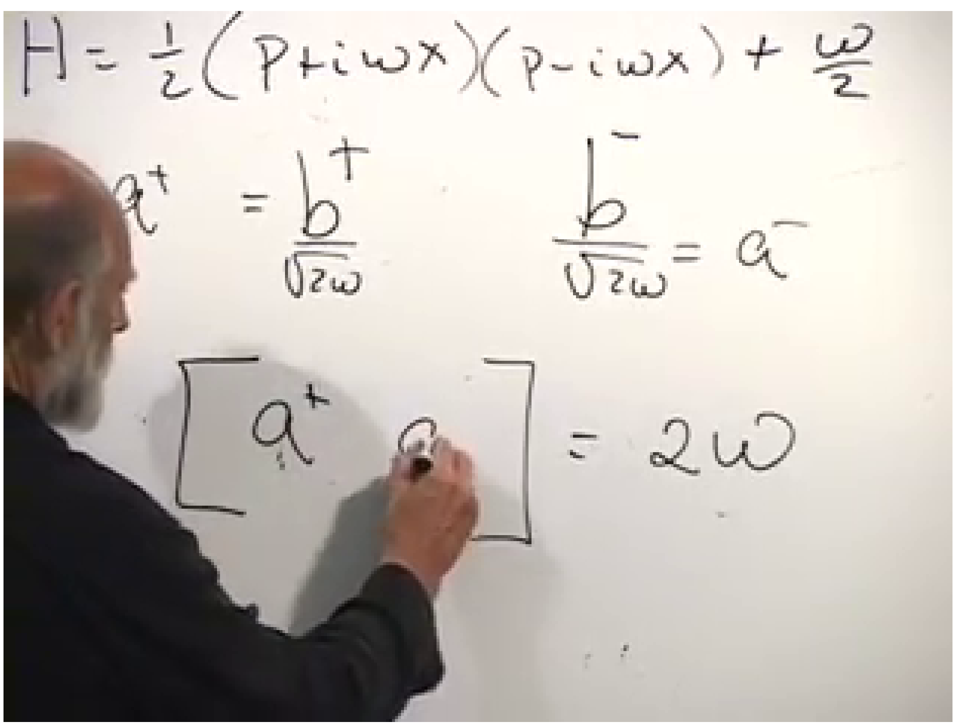

Figure 1.15: Rescaling the first-order factors produces dimensionless ladder operators with a unit commutator.

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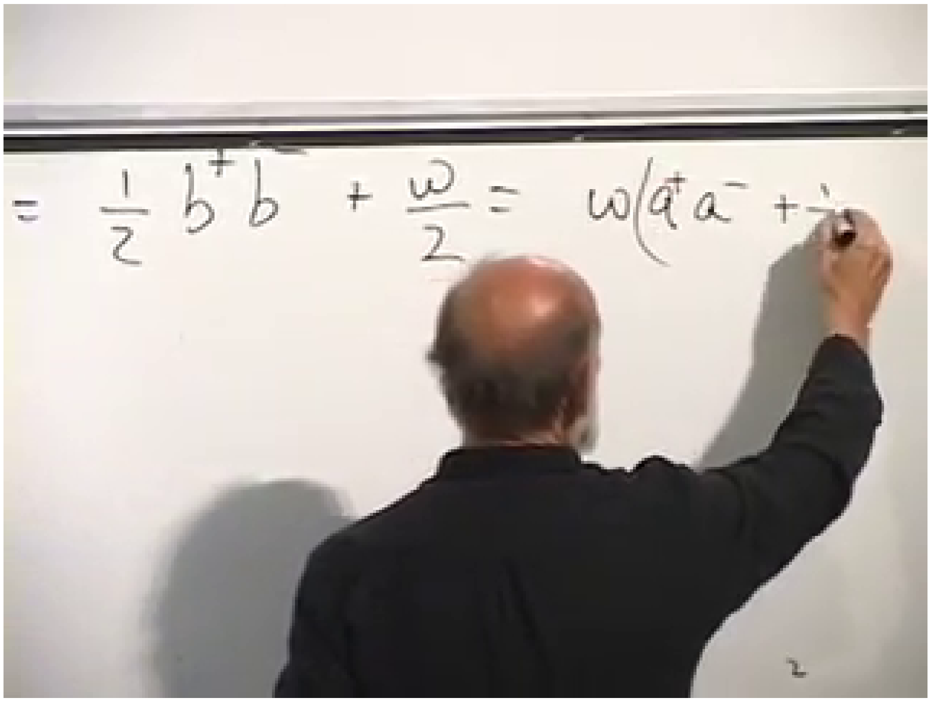

Figure 1.16: The oscillator Hamiltonian reduced to the number operator plus the zero-point half.
Lecture frame: 01:09:37.

The Hamiltonian is now

$$\boxed{\hat{H} = \hbar\omega\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right).} \tag{1.42}$$

The operator

$$\hat{N} = \hat{a}^{\dagger}\hat{a} \tag{1.43}$$

is positive:

$$\langle\psi|\,\hat{N}\,|\psi\rangle = \|\hat{a}\,|\psi\rangle\,\|^2 \geq 0. \tag{1.44}$$

This factorization proves, rather than merely guesses, that no state lies below $\hbar\omega/2$.

Question from the audience. Which commutator has sign $+1$, and which operator really annihilates the ground state? [a]

Response. With the consistent definitions in Equation (1.39), $\hat{a} \propto \hat{p} - i\omega\hat{x}$ annihilates the Gaussian, $[\hat{a}, \hat{a}^{\dagger}] = 1$, and therefore $[\hat{a}^{\dagger}, \hat{a}] = -1$. The extended board discussion mixed the naming order, not the physics. Testing the operator directly on ψ_0 fixes the convention unambiguously.

[a] Lecture timestamp: 01:11:05–01:14:09.

Writing the ground state as $|0\rangle$,

$$\hat{a}\,|0\rangle = 0, \qquad \hat{N}\,|0\rangle = 0, \qquad \hat{H}\,|0\rangle = \frac{1}{2}\hbar\omega\,|0\rangle\,. \tag{1.45}$$

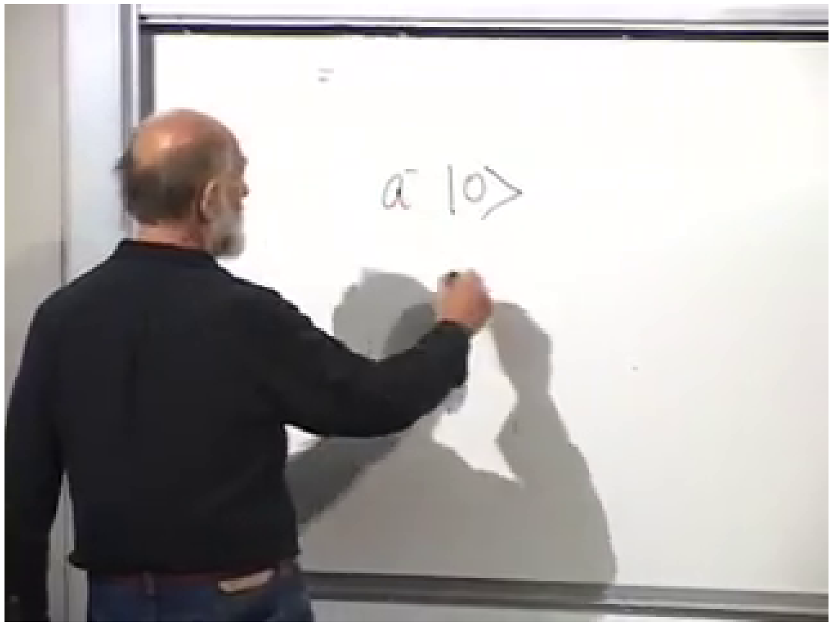

Figure 1.17: The algebraic definition of the ground state: $\hat{a}\,|0\rangle = 0$.
Lecture frame: 01:14:47.

1.7 Climbing the Energy Ladder

Suppose

$$\hat{N}\,|n\rangle = n\,|n\rangle\,. \tag{1.46}$$

Use the commutator to move $\hat{N}$ past the raising operator:

$$\begin{aligned}\hat{N}\hat{a}^\dagger &= \hat{a}^\dagger\hat{a}\hat{a}^\dagger &(1.47)\\ &= \hat{a}^\dagger\left(\hat{a}^\dagger\hat{a}+1\right) &(1.48)\\ &= \hat{a}^\dagger(\hat{N}+1). &(1.49)\end{aligned}$$

Acting on $|n\rangle$,

$$\hat{N}(\hat{a}^\dagger\,|n\rangle) = (n+1)\hat{a}^\dagger\,|n\rangle\,. \tag{1.50}$$

Thus $\hat{a}^\dagger\,|n\rangle$ is an eigenstate with number eigenvalue $n+1$, unless it vanishes.

The first excited state is proportional to $\hat{a}^\dagger\,|0\rangle$ and has energy

$$E_1 = \hbar\omega\left(1+\frac{1}{2}\right) = \frac{3}{2}\hbar\omega. \tag{1.51}$$

Repeating the argument generates

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\,|0\rangle\,, \qquad n = 0,1,2,\ldots, \tag{1.52}$$

with the normalized ladder actions

$$\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle\,, \qquad \hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle\,. \tag{1.53}$$

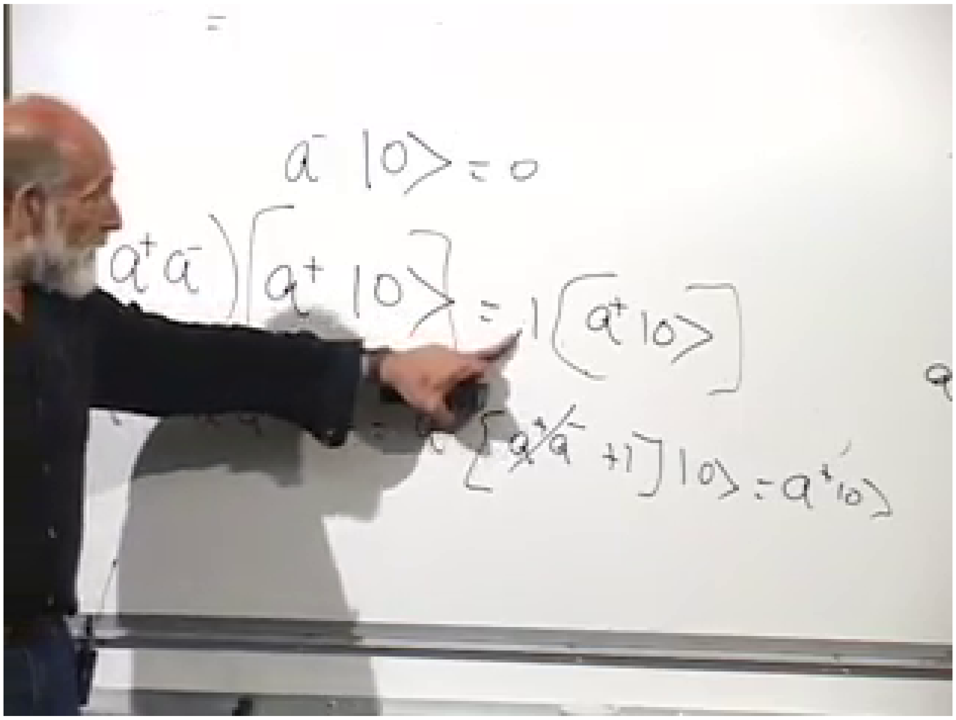

Figure 1.18: Commuting the number operator through $\hat{a}^\dagger$ raises its eigenvalue by one.
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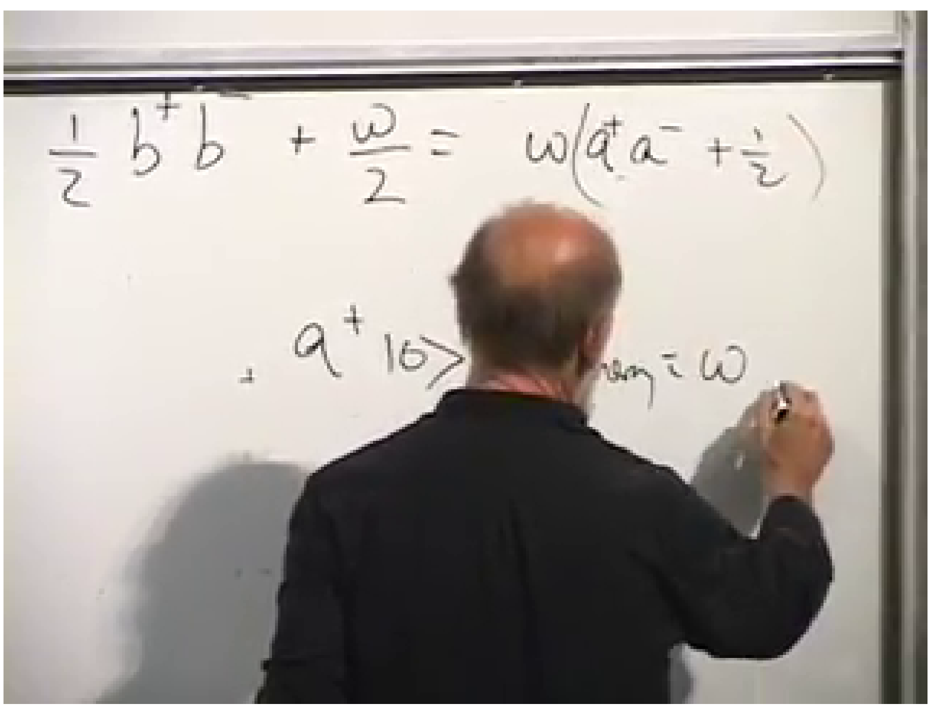

Figure 1.19: One application of the raising operator adds $\hbar\omega$ to the ground-state energy.
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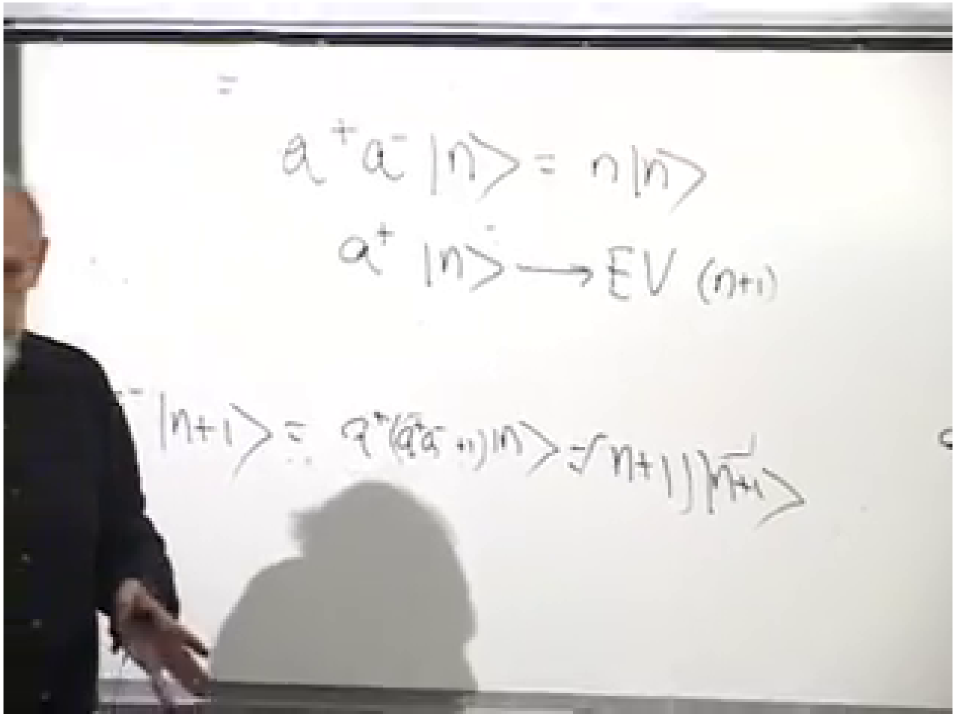

Figure 1.20: The raising theorem maps a number eigenstate to the next eigenstate.
Lecture frame: 01:24:52.

At the bottom, $\hat{a}\,|0\rangle = 0$, so the ladder cannot continue to negative n.

The complete spectrum is

$$\boxed{E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots.} \tag{1.54}$$

Successive levels are separated by the fixed quantum $\hbar\omega$:

$$E_{n+1} - E_n = \hbar\omega. \tag{1.55}$$

Question from the audience. Are there other Hamiltonians with an equally spaced spectrum? [a]

Response. One can construct them. For example, on a particle-on-a-ring Hilbert space, an operator proportional to angular momentum has equally spaced eigenvalues $m\hbar$ with positive and negative integers m. Used directly as a Hamiltonian, however, that spectrum is unbounded below and has no stable ground state. The oscillator is distinguished by its equally spaced ladder together with a bottom.

[a] Lecture timestamp: 01:27:35–01:28:54.

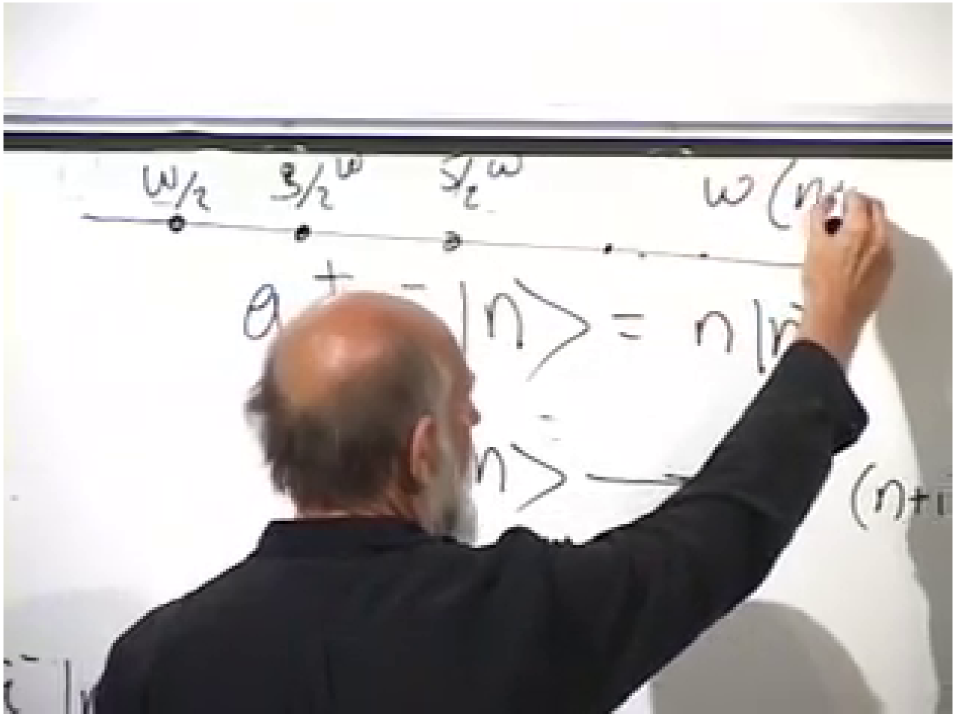

Figure 1.21: The equally spaced oscillator spectrum beginning at the zero-point energy.
Lecture frame: 01:25:38.

1.8 What the Eigenfunctions Mean

Question from the audience. What physical system does a quantum harmonic oscillator represent? Is it an electron orbiting an atom? [a]

Response. A Coulomb orbit is not itself a harmonic oscillator. The general setting is any stable configuration examined through small displacements. A diatomic molecule vibrating about its equilibrium bond length is a good example. More broadly, Equation (1.2) shows why every ordinary nondegenerate minimum is approximately quadratic. A quantum system is characterized by measurable probability distributions and spectra, not by a literal photograph of a microscopic trajectory.

[a] Lecture timestamp: 01:29:02–01:32:31.

For the ground state,

$$|\psi_0(x)|^2 = \sqrt{\frac{\omega}{\pi\hbar}}\exp\left(-\frac{\omega x^2}{\hbar}\right). \tag{1.56}$$

Its variance is

$$(\Delta x)^2 = \frac{\hbar}{2\omega} \tag{1.57}$$

in the unit-mass coordinate. Increasing ω stiffens the restoring force and narrows the spatial distribution.

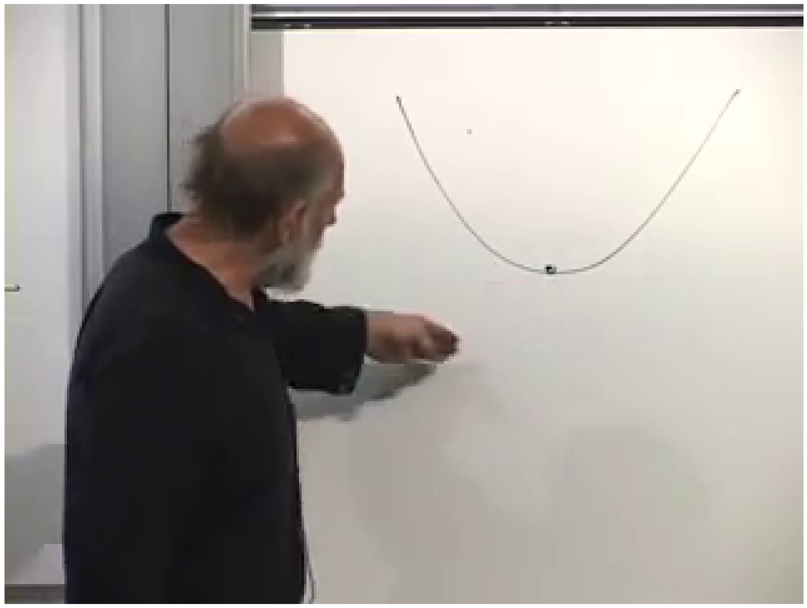

Figure 1.22: Near a stable minimum, a general potential is locally parabolic and therefore harmonic to leading order.

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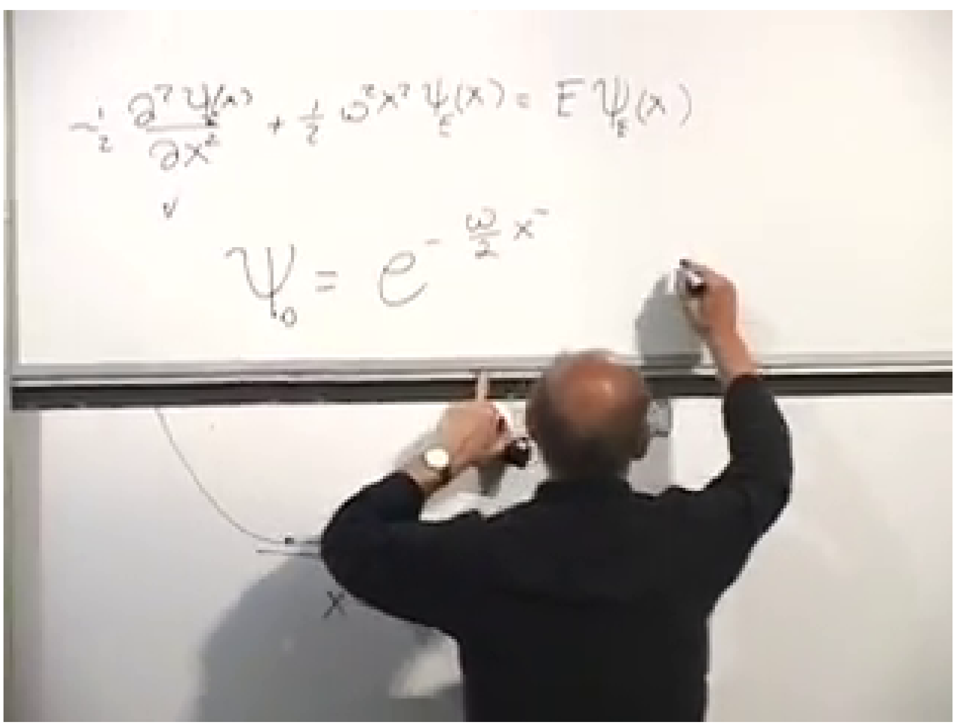

Figure 1.23: The Gaussian ground-state probability is more tightly localized for a stiffer oscillator.

Lecture frame: 01:32:38.

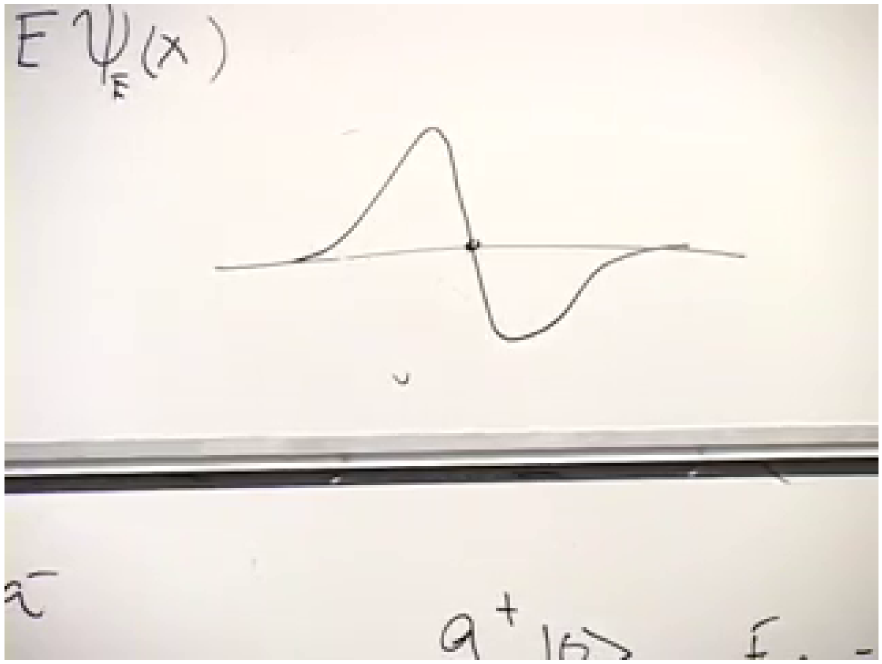

Figure 1.24: The first excited state is an odd Gaussian times x, with one node at the origin.
Lecture frame: 01:37:30.

Apply the raising operator once:

$$\psi_1(x) \propto \left(-\hbar\frac{d}{dx} + \omega x\right) e^{-\omega x^2/(2\hbar)} \tag{1.58}$$

$$\propto x\, e^{-\omega x^2/(2\hbar)}. \tag{1.59}$$

It is odd, changes sign at the origin, and has one node. Multiplying a state by an overall constant or phase, including a minus sign, does not change its physical predictions.

Repeated raising produces a polynomial times the same Gaussian:

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{\omega}{\pi\hbar}\right)^{1/4} H_n\left(\sqrt{\frac{\omega}{\hbar}}x\right) e^{-\omega x^2/(2\hbar)}, \tag{1.60}$$

where H_n is the nth Hermite polynomial. The parity alternates,

$$\psi_n(-x) = (-1)^n \psi_n(x), \tag{1.61}$$

and ψ_n has n real nodes. Higher states oscillate more rapidly, carry larger characteristic momentum, and extend farther from the origin, as expected for a more energetic oscillator.

Question from the audience. Are the nodes of the Hermite-polynomial wave functions at real values of x, and is the probability exactly zero there? [a]

Response. Yes. The nth oscillator eigenfunction has n simple real zeros. At each node $\psi_n(x) = 0$, so the position probability density $|\psi_n(x)|^2$ also vanishes. The zeros arise from destructive interference within the stationary wave function.

[a] Lecture timestamp: 01:41:44–01:42:58.

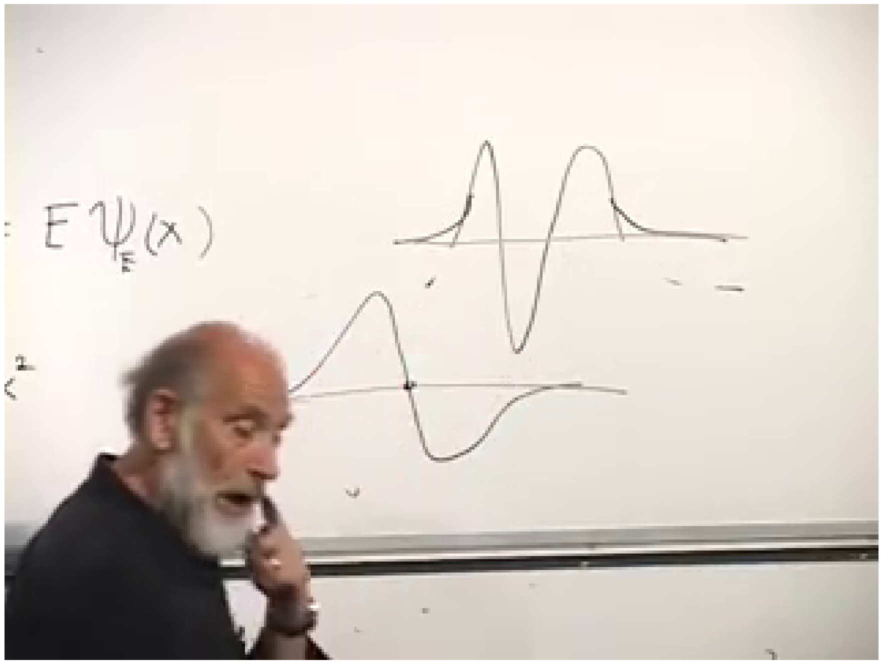

Figure 1.25: Successive oscillator states acquire additional nodes while retaining Gaussian decay at large distance.

Lecture frame: 01:39:12.

Question from the audience. What is the physical reason for quantized energy rather than a continuum? [a]

Response. The mathematical and physical statements are the same boundary condition viewed from two sides. The oscillator is confined: its potential grows without bound, so a physical eigenfunction must decay at both infinities. Those two boundary requirements can be met only at discrete energies. A free particle can escape to infinity and therefore has a continuous momentum and energy spectrum. More general potentials may have discrete levels below an escape threshold and a continuum above it.

[a] Lecture timestamp: 01:42:59–01:44:35.

Question from the audience. Why are the exponentially growing solutions excluded rather than treated as unusual states? [a]

Response. They cannot be normalized to a probability distribution on the line. Relative to their divergent weight at infinity, every finite region would have zero probability, and the oscillator potential would give them divergent energy. They are mathematical solutions of the local differential equation, but not state vectors in the oscillator Hilbert space.

[a] Lecture timestamp: 01:44:38–01:46:24.

1.9 The Completed Oscillator Picture

The harmonic oscillator begins with the most ordinary classical motion: $\ddot{x} = -\omega^2 x$. Quantization turns its coordinate and momentum into noncommuting operators. That single change prevents zero energy, selects a normalizable Gaussian ground state, and leaves a half-quantum of irreducible energy.

Factorization then converts a differential-equation problem into an algebra problem:

$$[\hat{a}, \hat{a}^\dagger] = 1, \qquad \hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right). \tag{1.62}$$

The creation operator climbs an equally spaced spectrum, the annihilation operator descends it, and the ground state terminates the descent. In position space the same ladder appears as Gaussian wave functions multiplied by successively higher Hermite polynomials.

Angular momentum has not been covered here; it is deferred rather than silently omitted. The oscillator itself supplies the bridge to the next subject, because every normal mode of a quantum field behaves as a quantum harmonic oscillator, and its raising operator creates one additional quantum of excitation.